SWAS KAUSHAL
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Research interest

I integrate quantitative genetics, plant breeding, and data-driven approaches to improve crop performance and resilience. My research combines high-throughput phenotyping (UAV-based), genomics, and machine/deep learning to model complex traits and genotype × environment interactions for more accurate selection decisions. I focus on developing scalable, field-based breeding strategies that integrate genomic, phenomic, and environmental data to accelerate genetic gain and enable sustainable, climate-resilient crop improvement.

Education

Ph.D. Plant Science 2022 - 2026
South Dakota State University,
Brookings, South Dakota, United States
GPA: 3.9/4

Graduate Certificate in Data Science 2023 - 2024
South Dakota State University,
Brookings, South Dakota, United States

M.S. Plant Breeding and Genetics 2018 - 2020
Punjab Agricultural University,
Ludhiana, India
GPA: 3.57/4

B.S. Agriculture (Hons.) 2014 - 2018
Lovely Professional University,
Phagwara, India
GPA: 3.99/4

Professional Experience

1. **Graduate Research Assistant**
   Winter Wheat Breeding Program, South Dakota State University (2022-2026)
   - Developed deep learning models including convolutional and attention-based architectures to predict grain yield using UAV-derived spectral and textural features across multiple environments and growth stages.

- Integrated genomic, phenomic, and environmental datasets to model complex traits such as yield, grain protein content, and disease resistance, enabling data-driven selection decisions.
- Established high-throughput phenotyping pipelines for UAV data processing and time-series analysis, improving efficiency and consistency of large-scale field evaluations.
- Designed and implemented multi-trait genomic selection models incorporating phenomic predictors to enhance selection accuracy for key agronomic and quality traits.

2. **Field Crop Breeding Phenomics Co-op**
   Syngenta, Minnesota (May 2025-December 2025)
   - Collected, processed, and analyzed high-throughput phenotypic data from large-scale soybean, corn, and hybrid wheat trials using UAV, satellite imagery, NIR sensors, and digital imaging platforms.
   - Integrated multi-source phenomic datasets to develop predictive models supporting early selection, line advancement, and product placement decisions in breeding pipelines
   - Applied data-driven approaches to enhance selection efficiency and collaborated with cross-functional breeding teams to translate phenomics insights into actionable outcomes.
3. **M.S. Research Experience**
   Punjab Agricultural University, India (2018-2020)
   - Conducted molecular and genetic characterization of irradiated introgression lines to identify genomic variation and assess stability of "B" genome segments in Brassica napus.
   - Applied molecular marker and genomic approaches to detect introgressed regions and evaluate their potential contribution to trait variation and breeding utility.
   - Analyzed genetic diversity and structural variation to support introgression breeding strategies and broaden the genetic base of breeding populations
4. **Product Marketing and Assessment Intern**
   Agronxt, Chandigarh (2017)
   - Conducted market research and field-level assessments to understand farmer needs and support development of data-driven product strategies for climate-smart agricultural solutions.
   - Contributed to product positioning and outreach by aligning soil testing and digital decision-support tools with stakeholder requirements, enhancing adoption and on-farm impact.

Skills

1. **Digital Phenotyping and Precision Agriculture**
    - Proficient in UAV-based high-throughput phenotyping (FAA Part 107 certified), including acquisition and processing of RGB and multispectral imagery under field conditions.
    - Experienced in geospatial and image analysis using Pix4D, ENVI, and GIS platforms to generate vegetation indices, canopy metrics, and spatial variability maps.
    - Developed standardized pipelines and protocols for scalable phenotyping workflows supporting breeding trials and field evaluations.

2. **Data Science Quantitative Genetics, and AI**
    - Strong programming skills in Python and R for statistical modeling, genomic analysis, and machine/deep learning applications in plant breeding.
    - Experienced with TensorFlow, PyTorch, Scikit-learn, and OpenCV for predictive modeling, image analysis, and multi-modal data integration.
    - Integrated genomic, phenomic, and environmental datasets to model complex traits and genotype × environment interactions for data-driven selection.
3. **Plant Breeding and Field Research**
    - Designed, implemented, and managed field and greenhouse experiments across breeding nurseries, including trial establishment, phenotyping, and harvest operations.
    - Applied advanced experimental designs and statistical analyses to optimize trial efficiency and accuracy.
    - Maintained high-quality field data, conducted soil and sample assessments, and supported selection, advancement, and evaluation decisions in applied breeding programs.
4. **Tools and Platforms**
    - ENVI, ArcGIS, ArcMap, Pix4D, GMATA, GGT, BioEdit, Geneious, CropStat, ICT, DARwin, TASSEL

Honors, Awards, and Scholarships

- Brookings Area Master Gardeners Scholarship, $1,000 (2026)
- Wheat Initiative Phenotyping Early Working Group Travel Award, European Plant Phenotyping Training School, Bonn, Germany, $1,800 (2025)
- Early Career Travel Award, National Association for Plant Breeding, Hawaii, $1,000 (2025)
- North American Plant Phenotyping Network Conference Travel Award, Johnston, IA, $1,000 (2025)
- University of Illinois Plant Science Symposium Travel Award, $1,000 (2024)
- Nebraska Plant Science Symposium Travel grant award, $750 (2024)
- AI in Agriculture and Natural Resources Travel grant award, $1000 (2024)

- AG2PI Travel Award, $1,500 (2024)
- Edgar S. McFadden Graduate Student Scholarship, South Dakota State University, $5,500 (2023)
- Encompass Scholar, Crop Science Society of America (2023)
- SDEPSCOR Science Communication Fellowship, South Dakota Discovery Center (2023)
- Gerald O. Mott Award, Crop Science Society of America (2023)
- International Arid Lands Consortium Program Fund, $1,000 (2022)
- Dr. Sukhdev Singh Award for Best Essay, Punjab Agricultural University, India (2019)
- University Merit Award, Punjab Agricultural University, India (2018, 2019)
- University Merit Fellowship for Masters, Punjab Agricultural University (2018)
- Second Place, Paper Presentation, Jalandhar, India (2016)
- University Merit Award, Lovely Professional University, India (2014, 2015, 2016 and 2017)

Teaching Experience

- Teaching assistant, Plant Breeding (PS746), Dr Melanie Caffe
  South Dakota State University, Fall 2024
  - Led hands-on sessions on genome-wide association studies (GWAS) and genomic selection, integrating quantitative genetics concepts with practical data analysis
  - Delivered lecture on genomic selection, emphasizing its application in modern breeding programs and decision-making.
- Teaching assistant, Advanced Plant Breeding (PS787) Dr Sunish Sehgal
  South Dakota State University, Spring 2025
  - Facilitated practical training on mixed models, quantitative genetic analysis, and genomic prediction for graduate-level students
  - Delivered lecture on high-throughput phenotyping and its integration into breeding pipelines.

Publications

- **Kaushal, S*.,** Gill, H.S., Billah, M.M., Khan, S.N., Halder, J., Bernardo, A., Amand, P.S., Bai, G., Glover, K., Maimaitijiang, M. and Sehgal, S.K., 2024. Enhancing the potential of phenomic and genomic prediction in winter wheat breeding using high-throughput phenotyping and deep learning. Frontiers in Plant Science, 15, p.1410249. https://doi.org/10.3389/fpls.2024.1410249
- **Kaushal, S*.,** Billah, M.M., Halder, J., Maimaitijiang, M. and Sehgal, S.K., 2026. High-throughput estimation of wheat tiller density using UAV multispectral imagery and attention-based deep learning. (In review, BMC Plant biology)

- **Kaushal, S*.,** Billah, M.M., Halder, J., Maimaitijiang, M. and Sehgal, S.K., 2026. Deep transfer learning for grain yield prediction using UAV-derived spectral and textural indices across temporal and spatial domains. (In review, The Plant Phenome)
- Billah, M.M., Maimaitijiang, **Kaushal, S**., Millet, B., Khan, S.N., Halder, J., Kleinjan, J., Sehgal, S.K., 2026. Predicting Wheat Yield and Grain Quality with UAV Multispectral Imagery and Deep Learning. (In review, Frontiers in Plant Science)
- Ghimire, H., Maimaitijiang, M., **Kaushal, S**., Koupal, D., Poudel, K., Thapa, S., Subedi, S., Halder, J. and Sehgal, S.K., 2026. Wheat Spike and Spikelet Detection on Close-range Digital Imagery using Deep Learning. (In review, Plant Phenomics)
- Maimaitijiang, M., Ghimire, H., Thapa, S., Billah, M.M., Sehgal, S., Singh, M., **Kaushal, S**., Poudel, K., Subedi, S., Janjua, U.U.R. and Makonga, L.O., 2026. WheatAI v1. 0: An AI-Powered High Throughput Wheat Phenotyping Platform. arXiv preprint arXiv:2601.08863.
- Halder, J., Saini, DK., Gill, H., **Kaushal, S**., Sehgal, SK., 2025. Mapping spike and kernel traits in wheat using a recombinant Inbred population. (In preparation)
- **Kaushal, S*.** 2020. Molecular-genetic characterization of irradiated 'B' genome introgression lines of Brassica Napus. M.Sc. Plant Breeding and Genetics, Thesis, PAU-Punjab Agricultural University, Ludhiana, India.

<u>Conference/Symposium Presentations</u>

- **Kaushal, S**., Billah, M.M., Halder, J., Maimaitijiang, M. and Sehgal, S.K., 2026. Leveraging AI and high throughput phenotyping to bridge genotype, phenotype, and environment scales for yield and component traits in winter wheat, Plant and Animal Genome, California (Oral)
- **Kaushal, S**., Billah, M.M., Halder, J., Maimaitijiang, M. and Sehgal, S.K., 2025. Optimizing winter wheat breeding with high-throughput phenotyping and deep transfer learning, WheatCAP Annual Meeting (Oral)
- **Kaushal, S** and Sehgal, S.K., 2025. artificial intelligence and machine learning for predictive breeding and genetic advancement, Punjab Agricultural University, India (Invited Lecture)
- **Kaushal, S**., Billah, M.M., Halder, J., Maimaitijiang, M. and Sehgal, S.K., 2025. UAV-based deep transfer learning to improve grain yield prediction in winter wheat across temporal, spatial, and phenological variability, National Association of Plant Breeding, Hawaii (Poster)
- Ghimire, H., Maimaitijiang, M., **Kaushal, S.,** Koupal, D and Sehgal, S.K., 2024. Wheat head detection from close-range digital imagery using deep learning, Great Plains and Rocky Mountain Annual meeting, Orem, Utah (Poster)
- **Kaushal, S**., Billah, M.M., Halder, J., Maimaitijiang, M. and Sehgal, S.K., 2024. Deep transfer learning for grain yield prediction using UAV-derived spectral and textural indices across temporal and spatial domains, UIUC, Urbana-Champaign (Oral)

- **Kaushal, S**., Billah, M.M., Halder, J., Maimaitijiang, M. and Sehgal, S.K., 2024. Deep Learning and high-throughput phenotyping: advancing winter wheat breeding, AI in Agriculture and Natural Resources Conference, A&M, Texas (Oral)
- **Kaushal, S**., Gill, H., Halder, J., Billah, MM., Shahid, N., Maimaitijiang, M., Bernardo, A., St Amand, P., Bai, G., Sehgal, SK., 2024. Improving grain yield prediction with multi-modal deep learning, integrating genomics and phenomics, NAPPN, West Lafayette, Indianapolis (Oral)
- **Kaushal, S**., Gill, H., Halder, J., Billah, MM., Shahid, N., Maimaitijiang, M., Bernardo, A., St Amand, P., Bai, G., Sehgal, SK., 2024. AI-driven winter wheat phenomic prediction utilizing UAV multispectral data, WheatCAP Annual Meeting, California (Oral)
- **Kaushal, S**., Gill, H., Halder, J., Billah, MM., Shahid, N., Maimaitijiang, M., Bernardo, A., St Amand, P., Bai, G., Sehgal, SK., 2023. Evaluating the potential of genomic and phenomic selection in hard winter wheat breeding program**,** Plant and Animal Genome, California (Poster)
- Billah, MM., **Kaushal, S**., Shahid, N., Kleinjan, J., Millett, B., Maimaitijiang, M., Sehgal, SK., 2023. AI-driven wheat yield and protein content forecasting using UAV remote sensing, SDSU Data Science Symposium, South Dakota (Poster)
- Thapa, S., **Kaushal, S**., Maimaitijiang, M., Halder, J., Rana, A., Nafchi, A., Shaukat, A., Sehgal, SK., 2022. Estimating FHB infection level in winter wheat spikes using high-resolution imaging and deep transfer learning, National FHB Forum Annual Meeting, Florida (Poster)
- Billah, MM., **Kaushal, S**., Shahid, N., Kleinjan, J., Millett, B., Maimaitijiang, M., Sehgal, SK., 2022. Estimating winter wheat yield and protein using UAV multispectral imagery and machine learning, AAG Great Plains-Rocky Mountain Division Annual Meeting, Colorado (Poster)
- **Kaushal, S**., Shahid, N., Rana, A., Thapa, S., Halder, J., Maimaitijiang, M., Sehgal, SK., 2022. Deploying machine learning for selection of fusarium head blight resistance in wheat, NAPB Annual Meeting, Iowa (Poster)
- **Kaushal, S**., Shahid, N., Rana, A., Halder, J., Maimaitijiang, M., Sehgal, SK., 2022. Deploying machine learning for selection of fusarium head blight resistance in wheat, Gamma Sigma Delta Honor Society of Agriculture, South Dakota (Poster)

Extension and Outreach

- Delivered a UAV-based phenotyping demonstration at the Dakota Lakes Research Farm Field Day (June 2022), showcasing applications of high-throughput phenotyping for improving selection efficiency in winter wheat breeding to producers and stakeholders.
- Presented at the 33rd Annual Draper Winter Wheat Meeting (August 2022), communicating the role of UAV-based phenomics in enhancing trait evaluation and decision-making in applied breeding programs.

Grants

Funded

- Developing a wheat yield early forecasting system using ground & satellite imaging and artificial intelligence, SD Wheat Commission, contributed to proof of concept, data collection, and writing, $25,000 (2024)

Submitted

- AI-driven field-based high-throughput phenotyping integrating genomic, phenomic, and enviromics data for transferable modeling, Jackrabbit Jumpstart Grant Initiative,$25,000 (2024)
- YieldView: AI-based wheat yield early forecasting tool leveraging synergistic application of ground and satellite data to support in-season field management, Student PI, NCR-SARE, $15,000 (2023)
- AI-based prediction of yield in wheat variety development under no-till system, Student PI, NCR-SARE, $15,000 (2022)

Mentoring and Supervision

- **UAV-based phenotyping and data acquisition in winter wheat breeding**, Mentored Conner Thaler (2021-2022) and Dante Koupal (2022-2025) in UAV operations, flight planning, and processing of field-based phenomic data.
- **Field imaging, LAI, and biomass estimation using high-throughput phenotyping**, Mentored Mackenzie Henning (2022-2024) in image-based trait extraction and analysis for breeding applications.
- **Wheat nutritional and quality trait analysis (arabinoxylans),** Mentored Pradeep Sitaula (2024-2026) on physiological and biochemical characterization of grain quality traits and their implications for end-use quality and nutritional value.
- **Field operations and breeding nursery management**, Supervised teams of 8-10 members (202-2025) in planting, phenotyping, harvesting, and seed processing for winter wheat breeding programs.
- **Extension and stakeholder engagement through field days**, Contributed to organization and outreach activities (2022–2025) to communicate breeding advancements and phenotyping technologies to producers.

Workshops and Professional Training

- Science Communications Course, Something Else, WheatCAP (2026)
- Tools for Data Science, IBM (Coursera, 2026)
- What is Data Science, IBM (Coursera, 2025)
- DSSAT, University of Georgia, Atlanta, USA (2023)
- Basic Wheat Improvement Course, CIMMYT (2022)
- Agricultural Crop Classification with Synthetic Aperture Radar and Optical Remote Sensing training series, NASA (2021)

Leadership

- Chair, Graduate Student Leadership Committee, ASA, CSSA, SSSA (2026)
- Member, CSSA Graduate Student Committee, ASA, CSSA, SSSA (2026)
- Member, CSSA Graduate Student Committee, ASA, CSSA, SSSA (2025)
- Member, Graduate Student Leadership Committee, ASA, CSSA, SSSA (2025)
- Member, CSSA Fellows committee (2025)
- Co-chair, "From AI to Ag: Using Data-Driven Decisions to Inform the Future of Crop Improvement" (2025)
- President, Plant Science Graduate Student Association (PSGSA) (2023-2024)
- Treasurer, Plant Science Graduate Student Association (PSGSA) (2022-2023)
- President, SDSU Cricket Club (2023-24)

Professional Memberships

- Crop Science Society of America (CSSA), 2021-present
- American Society of Agronomy (ASA), 2021-present
- National Association for Plant Breeding (NAPB), 2022-present
- North American Plant Phenotyping Network (NAPPN), 2022-present

References

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